

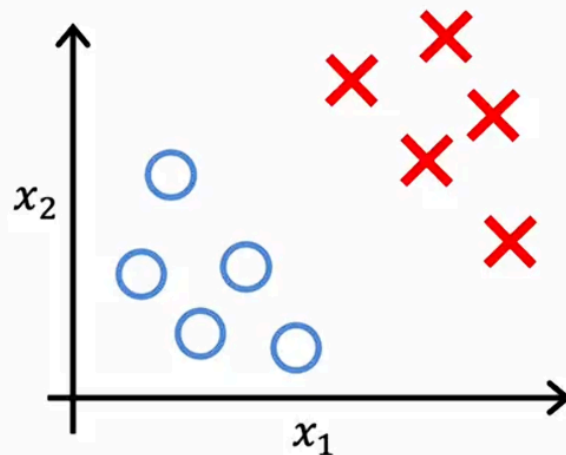
Day 28. Clustering

Clustering

What is Clustering?

Recall on supervised learning

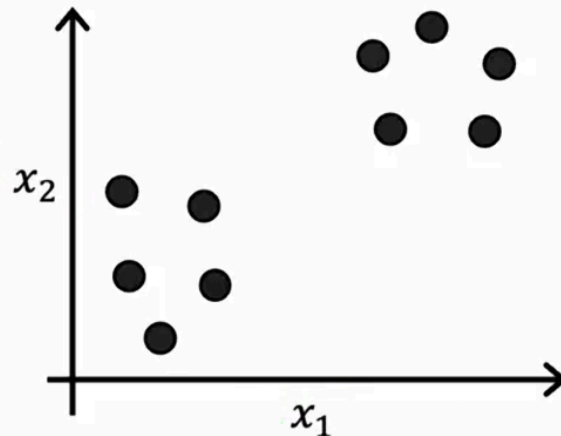
Supervised learning



Training set: $\{(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), (x^{(3)}, y^{(3)}), \dots, (x^{(m)}, y^{(m)})\} ?$

however, in unsupervised learning, our data is something like this.

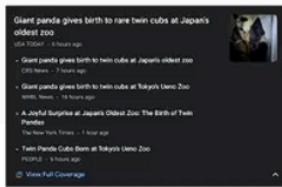
Unsupervised learning



Training set: $\{x^{(1)}, x^{(2)}, x^{(3)}, \dots, x^{(m)}\}$

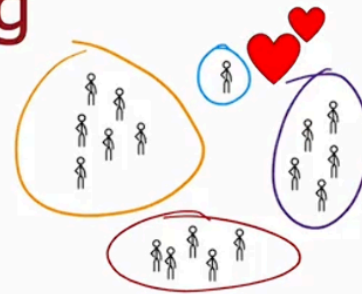
we want the algorithm to find interesting patterns from the dataset.

Applications of clustering

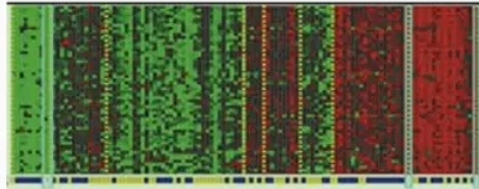


Grouping similar news

- Growing skills
- Develop career
- Stay updated with AI, understand how it affects your field of work



Market segmentation



DNA analysis



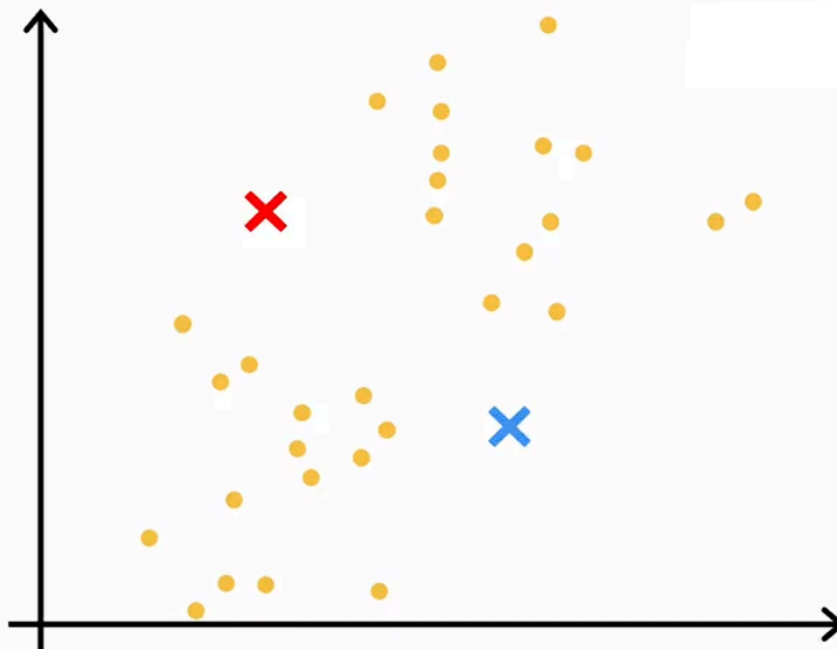
Image credit: NASA/JPL-Caltech/E. Churchwell (Univ. of Wisconsin, Madison)

Astronomical data analysis

K-means

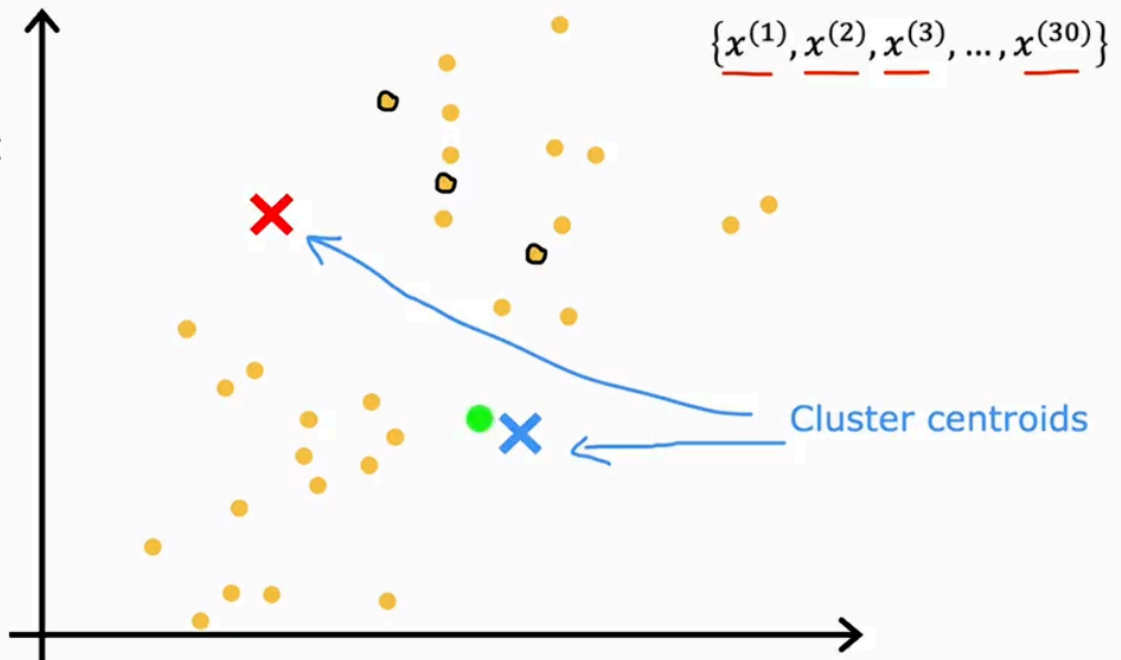
intuition

First we will tell the algorithm the number of clusters we want to find, then it will randomly guess the center of the clusters. (will discuss how to decide the number of clusters)

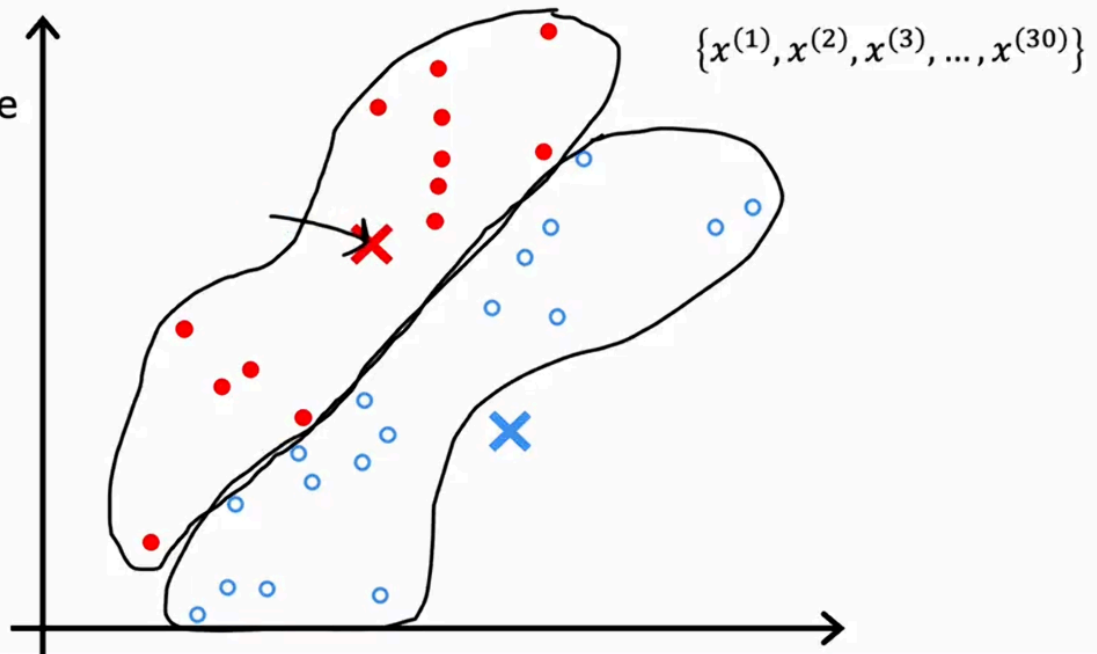


Step 1:

Assign
each point
to its
closest
centroid

**Step 2:**

Recompute
the
centroids



then we repeat step 1 and step 2, until we there's no further changes of the centroids' locations and the group of the points.

Algorithm

K-means algorithm

Randomly initialize K cluster centroids $\mu_1, \mu_2, \dots, \mu_K$

Repeat {

Assign points to cluster centroids

for $i = 1$ to m

$c^{(i)} :=$ index (from 1 to K) of cluster centroid closest to $x^{(i)}$

Move cluster centroids

for $k = 1$ to K

$\mu_k :=$ average (mean) of points assigned to cluster k

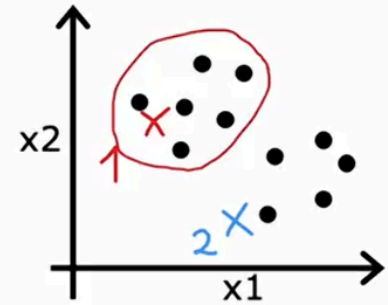
}

μ_1, μ_2

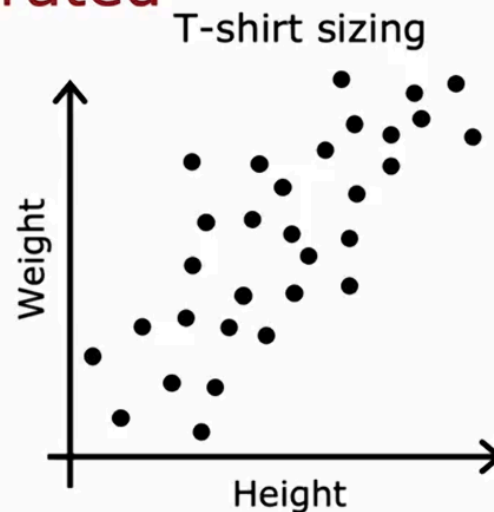
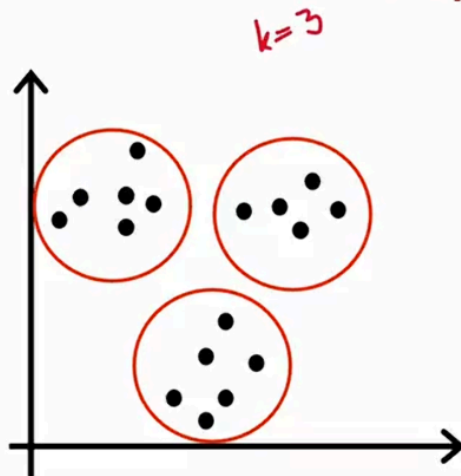
$x^{(1)}, x^{(2)}, \dots, x^{(n)}$

$n = 2$

$K = 2$



K-means for clusters that are not well separated



Optimization objective

K-means optimization objective

$c^{(i)}$ = index of cluster $(1, 2, \dots, K)$ to which example $x^{(i)}$ is currently assigned

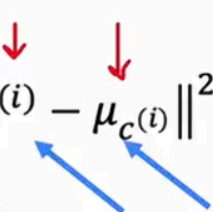
μ_k = cluster centroid k

$\mu_{c^{(i)}}$ = cluster centroid of cluster to which example $x^{(i)}$ has been assigned

Cost function

$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K) = \frac{1}{m} \sum_{i=1}^m \|x^{(i)} - \mu_{c^{(i)}}\|^2$$

$\min_{c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K} J(c^{(1)}, \dots, c^{(m)}, \mu_1, \dots, \mu_K)$



Handwritten red annotations: $x^{(10)}$, $c^{(10)}$, $\mu_c^{(10)}$

this function is called distortion function

moving the centroid to the average location is exactly the way to minimize this "cost" function.

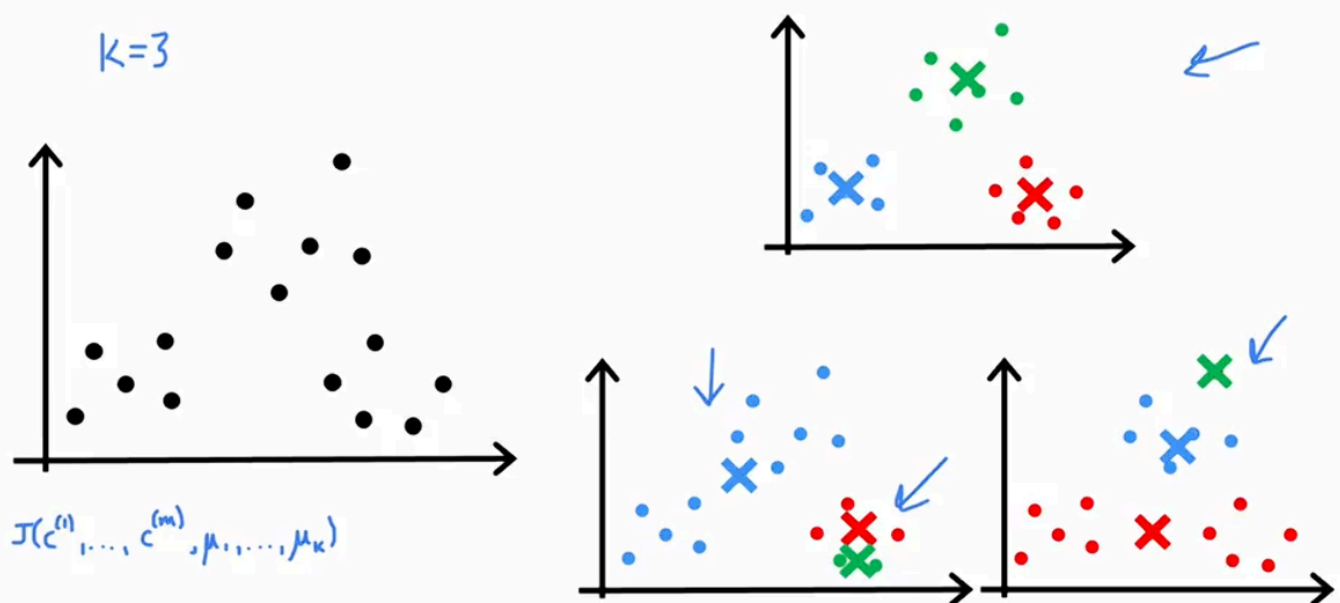
Initializing K-means

Random initialization

Choose $\underline{K} < \underline{m}$

Randomly pick K training examples.

Set $\mu_1, \mu_1, \dots, \mu_k$ equal to these K examples.



Random initialization

For $i = 1$ to 100 {

Randomly initialize K-means.

k random examples

Run K-means. Get $c^{(1)}, \dots, c^{(m)}, \mu_1, \mu_1, \dots, \mu_k$

Computer cost function (distortion)

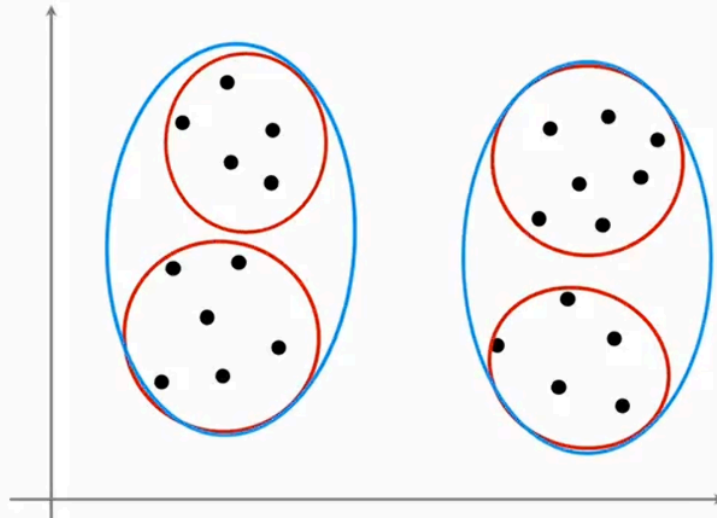
$$J(c^{(1)}, \dots, c^{(m)}, \mu_1, \mu_1, \dots, \mu_k)$$

}

pick set of clusters that gave lowest cost J

Choosing the numbers of clusters

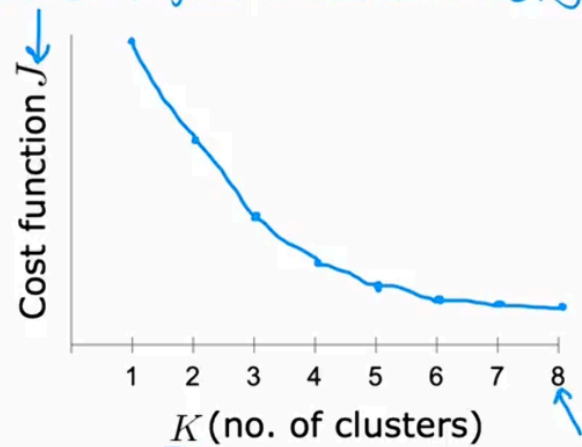
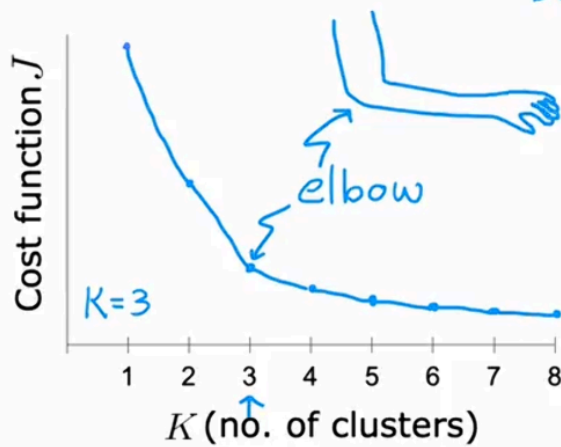
What is the right value of K?



Choosing the value of K

Elbow method

the right "K" is often ambiguous
Don't choose K just to minimize cost J



then how do we choose the cluster numbers?

Choosing the value of K

Often, you want to get clusters for some later (downstream) purpose.
Evaluate K-means based on how well it performs on that later purpose.

